Saanvi Reddy Baradi

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PROFESSIONAL SUMMARY

AI Engineer with 4 years of Python engineering experience across healthcare, payments, and banking, now building production Generative AI applications. Builds RAG pipelines and agentic workflows with LangChain, LangGraph, FastAPI, Azure OpenAI, and Azure AI Search, with a focus on retrieval quality, LLM evaluation (LangSmith, Ragas), and guardrails for sensitive data (PII). Brings a backend foundation from healthcare data, payments, and banking roles: REST APIs, SQL and PostgreSQL, data pipelines, testing, and CI/CD.

EDUCATION

University of South Florida, Tampa, FL | Master of Science, Artificial Intelligence and Business Analytics

WORK EXPERIENCE

AI Engineer | Meta, USA **Sep 2026 – Present**

- Develop FastAPI services for internal knowledge retrieval that return schema-checked, Pydantic-validated LLM outputs to downstream consumers.
- Tune chunking, embedding models, and FAISS index settings in LangChain to improve retrieval precision and cut irrelevant context passed to the LLM.
- Package retrieval, LLM calls, and output validation into reusable LangGraph components shared across the team's workflows.
- Add automated grounding, relevance, and consistency checks to PyTest so prompt and model changes are regression-tested before merge.
- Triage API and retrieval defects and review pull requests on the team's shared codebase.

AI Engineer | Truist Bank, Tampa, FL **Oct 2025 – Jul 2026**

- Built a multi-agent assistant on LangGraph, FastAPI, and Azure OpenAI that routes customer servicing, compliance, and operations queries to specialized agents for account services, transaction research, and banking products, cutting manual research time for operations users by about 20%.
- Developed RAG pipelines over internal financial documents in Azure Blob Storage, with Airflow-scheduled ingestion, chunking, Azure AI Search hybrid search, and reranking; improved retrieval relevance by about 12% on Ragas test sets.
- Built LangGraph agent workflows with intent classification, controlled tool calling, and retrieval for multi-step queries; contributed to MCP integrations exposing approved read-only tools with RBAC-scoped access and audit logging.
- Implemented LLM evaluation with LangSmith and Ragas on curated banking test sets to track faithfulness and relevance across releases, and added PII redaction and guardrails to block unsupported responses.
- Instrumented LLM latency, token usage, and retrieval failures with LangSmith and OpenTelemetry to pinpoint slow chains and failed retrievals before release.
- Wrote PyTest suites, Dockerized services, and deployed them to Azure Kubernetes Service (AKS) through Azure DevOps CI/CD pipelines; took part in code reviews and sprint planning.

Project Assistant | University of South Florida, Tampa, FL **Feb 2025 – Sep 2025**

- Built a Python ETL pipeline that standardized 180+ country-name variants to ISO codes using RapidFuzz and custom override rules, cutting mismatch errors by 90%.
- Automated ingestion, preprocessing, schema validation, and metadata generation for research datasets, reducing manual data preparation effort by 70%.
- Integrated the U.S. Census ACS 5-Year API with FIPS logic to pull demographic and income data by region.
- Kept pipelines in Git with CI checks and wrote usage notes so others on the project could rerun them.

Software Engineer, Python Backend | PayPal, Bengaluru, India **Jul 2023 – Jul 2024**

- Developed Python backend services and FastAPI REST APIs for internal payment operations and reporting.
- Wrote and optimized PostgreSQL queries for transaction lookup, reconciliation, and internal reporting workflows.
- Built reusable modules for transaction processing, business-rule validation, and structured logging.
- Wrote unit and integration tests with PyTest covering API responses, transaction logic, and failure scenarios.
- Supported AWS deployments through log analysis, configuration updates, and production troubleshooting.
- Wrote API documentation and worked through defects in sprint planning and code reviews.

Python Developer | IQVIA, Pune, India **Sep 2021 – Jun 2023**

- Developed Django REST API endpoints with request validation and database operations for healthcare applications.
- Wrote Python and Pandas scripts to clean and transform clinical trial data from CSV, Excel, and databases for reports.
- Wrote SQL queries and data validation checks to support reporting and verify the accuracy of healthcare datasets.
- Fixed assigned defects, wrote unit tests, and delivered small features with guidance from senior developers.
- Tested endpoints with Postman, maintained technical documentation, and helped with release verification.

PROJECTS

Agentic AI Platform | Python, LangGraph, FastAPI, pgvector, Ollama, Redis, Ragas, Docker

- Built a LangGraph assistant that routes queries between pgvector document RAG and a text-to-SQL path, returning cited answers with conversation context.
- Secured the SQL path with AST-based validation and read-only enforcement, plus JWT-authenticated FastAPI endpoints and Redis caching; reached 70% execution accuracy on a text-to-SQL test set.
- Wrote PyTest cases for the routing and SQL-validation logic to catch regressions when changing prompts or models.
- Ran Ragas evaluations in GitHub Actions CI, packaged the stack with Docker, Helm, and Terraform, and added OpenTelemetry tracing.

Fine-Tuned LLM Serving Pipeline | Mistral-7B, QLoRA, PEFT, Unsloth, Pinecone, FastAPI

- Fine-tuned Mistral-7B with 4-bit QLoRA and PEFT using Unsloth on a single T4 GPU.
- Cleaned and formatted the instruction dataset with Pandas, removing duplicate and malformed examples.
- Served it through a FastAPI RAG service with Pinecone retrieval, evaluated with Ragas and LLM-as-judge.
- Deployed the service to AWS EKS with Helm charts and Terraform, with GitHub Actions CI/CD for tests and builds.

RAG Document Q&A API | LangChain, FastAPI, pgvector, Vertex AI, Gemini

- Built a PDF question-answering API using Vertex AI embeddings, pgvector similarity search, and Gemini answers with source references.
- Added ingestion, health-check, and latency-tracking endpoints, PyTest API tests, and Docker packaging; deployed on Google Cloud Run with a live Hugging Face Spaces demo.
- Managed settings through environment variables and wrote a README with setup steps and example API calls.

TECHNICAL SKILLS

Languages: Python, SQL
Generative AI: RAG, Agentic Workflows (LangChain, LangGraph), Tool Calling, MCP, Prompt Engineering, Guardrails, Structured Outputs (Pydantic)
LLM Platforms & Evaluation: Azure OpenAI, Azure AI Search, Vertex AI, Gemini, Ollama, LangSmith, Ragas
Retrieval: pgvector, FAISS, Pinecone, Hybrid Search, Reranking
Fine-Tuning: QLoRA / LoRA with PEFT and Unsloth (project use)
Backend & Data: FastAPI, Django REST, PostgreSQL, Redis, Pandas, Apache Airflow, ETL Pipelines
Cloud & DevOps: Azure (AKS, Blob Storage, Azure DevOps), AWS, GCP (Vertex AI, Cloud Run), Docker, Kubernetes, GitHub Actions, CI/CD, OpenTelemetry; EKS, Helm, Terraform (project use)
Testing & Tools: PyTest, unittest, Postman, Git
Domains: Healthcare & Life Sciences (clinical data), Banking & Financial Services, Payments